

# Zhenrui Zheng

✉ ch4acko3@outlook.com • 📞 +86 159-7977-6837 • 🎓 Google Scholar Profile

## EDUCATION

### The Chinese University of Hong Kong, Shenzhen

- Ph.D. Candidate in Computer Science | Expected Graduation: June 2030 Sep 2025 – Present
- Advisor: Prof. Chenjun Xiao
  - Research Interests: Applying Machine Learning to Sequential Decision-Making Tasks (e.g., Reasoning), Generative Models and Reinforcement Learning. I am particularly driven by the pursuit of elegant, robust, and theoretically sound solutions to fundamental problems in research.

### Harbin Engineering University

- B.Eng. in Aircraft Design and Engineering Sep 2021 – Jun 2025
  - Weighted average score: 86.93/100 (Ranked 18/65)

## AWARDS & SCHOLARSHIPS

- **NeurIPS Competition 2024: AIGB Track Winner** (Final standing 4/793) Nov 2024  
Auto-Bidding in Large-Scale Auctions: Learning Decision-Making in Uncertain and Competitive Games  
Implement offline reinforcement-learning agents based on generative models to solve sequential decision-making and game problems in advertising. Compete with professional algorithm engineers from well-known companies/universities and get fourth place (Other listed teams: Kuaishou<sup>1st</sup>, BUAA<sup>2nd</sup>, ByteDance<sup>3rd</sup>, **My Team**<sup>4th</sup>, PKU<sup>10th</sup>, THU<sup>15th</sup>, Chinese Academy of Sciences<sup>16th</sup>, SNU<sup>22th</sup>)
- **The 2022 ICPC Asia Hangzhou Regional Contest: Silver Medal** Dec 2022  
The International Collegiate Programming Contest (ICPC) is an annual multi-tiered competitive programming competition among the universities of the world. Directed by ICPC Executive Director and Baylor Professor William B. Poucher, the ICPC operates autonomous regional contests covering six continents.
- **The GBA International Programming Contest: Third Prize** Apr 2025  
The GBA international programming contest is organized by the Shenzhen SLAI and The CUHKSZ, provides a platform for students to demonstrate their innovative abilities and skills in analyzing and solving problems.
- **The Chinese Mathematics Competitions: Provincial First Prize** Dec 2021  
National calculus competition for undergraduates. Obtained in the first semester of university, final standing 317/2300
- **Contemporary Undergraduate Mathematical Contest in Modeling: National Second Prize** Oct 2022  
One of 1,445 winning teams nationwide
- **Interdisciplinary Contest in Modeling: Meritorious Winner** Feb 2024
- **Collegiate Computer Systems & Programming Contest: Bronze Medal** Dec 2022
- **First-Class Scholarship for Outstanding Students of the School** May 2022
- **Zhou Peiyuan Mechanics Competition for College Students: Provincial Second Prize**
- **National Undergraduate Electronics Design Contest: Provincial Third Prize**

## RESEARCH EXPERIENCE

### Harbin Engineering University ACM/ICPC Team

Mar 2022 – Dec 2022

Represent the school in ACM/ICPC competitions, be responsible for relevant training courses within the school, and organize regional competitions.

### Harbin Engineering University Aircraft Control Laboratory

Dec 2022 – Nov 2023

Mainly research aircraft simulation, theory and application of advanced control algorithms, and application of statistical learning methods in the control field.

- Built a high-performance C++/Python-mixed aircraft simulation for the lab.
- Solve problems in discrete mathematics for the lab.

### Academy of Aerospace Solid Propulsion Technology

Jul 2023 – Oct 2023

Build a reinforcement learning environment framework and write high-performance simulation of aircraft aerodynamics. Training multi-agent reinforcement learning algorithms (MADDPG) in engineering projects.

### Tianjin University Deep Reinforcement Learning Laboratory

Nov 2023 – Present

A top research team in the field of reinforcement learning in China, focusing on cutting-edge research in the field of decision-making intelligence.

- Completed a paper as co-author and submitted to AAAI.
- As a core developer, formed a team to participate in a NeurIPS competition and won a top prize.

## ENGINEERING PROJECTS

### Swarm Aircraft Decision Intelligence Project

Jul 2023

Academy of Aerospace Solid Propulsion Technology

- Programming aircraft six degrees of freedom pneumatic physics simulation
- Responsible for the design, deployment and training of reinforcement learning algorithms
- Coordinate with other departments to complete joint programming

### Aerodrome: A Lightweight physics simulation platform for reinforcement learning

Jan 2025

C++/Python joint programming reinforcement learning platform, my undergraduate graduation project

- High-performance C++ aircraft dynamics simulation
  - Highly extensible code framework
  - Standardized reinforcement learning environment and Python interface

Github Repo: <https://github.com/CH4ACKO3/Aerodrome>

### ResearchFlow: ML research workflow management toolkit

Oct 2025

Python-based research toolkit for flexible process scheduling, model/data storage and management

- JSON-based indexed file storage
  - Retrieve model and result files by metadata querying, with several training parameter
  - Multi-process and crash-safe
- Asynchronous multi-GPU task scheduler
  - Dynamic computing resource monitoring and online GPU scheduling
    - Automatically suspend tasks when the system is critical to avoid crashes.
    - Withdraw tasks from a GPU and reassign at any time.
    - Fully asynchronous and low-latency file I/O, task pool management, compute device management
  - Interact with original text documents, researcher-friendly
  - Robust crash and interrupt handling
  - Comprehensive test suite with 54 test cases (unit, integration, concurrency)

Github Repo: <https://github.com/CH4ACKO3/researchflow>

## PUBLICATIONS

MODULI: Unlocking Preference Generalization via Diffusion Models for Offline Multi-Objective Reinforcement Learning. (*ICML2025*)

Yifu Yuan, **Zhenrui Zheng**, Zibin Dong, Jianye HAO.

Github Repo: <https://github.com/pickxiguapi/MODULI>

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